



UNIVERSITAT POLITÈCNICA DE CATALUNYA
BARCELONATECH

Centre de la Imatge i la Tecnologia Multimèdia

Introduction

1.2 Fundamentals of Networking

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Xarxes i Jocs On-line – 2025-2026
Grau en Disseny i Desenvolupament de Videojocs

Remember from last session



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- On-line games proliferated along with the emergence of personal computers and networking.
- In on-line games, the main challenge is to have a consistent state across players in an efficient and fair way.
- The network is complex and unreliable and will make this illusion hard to obtain; but we'll give you the tools to try to make it happen.

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- In on-line games, the main challenge is to have a consistent state across players in an efficient and fair way.
- The network is complex and unreliable and will make this illusion hard to obtain; but we'll give you the tools to try to make it happen.
- **Today we will explain the fundamental concepts and metrics behind any network that will help start understanding the complexity of the Internet.**

1. Introduction

- General aspects
- **Fundamentals of networking**
- Graph theory

2. The protocol stack

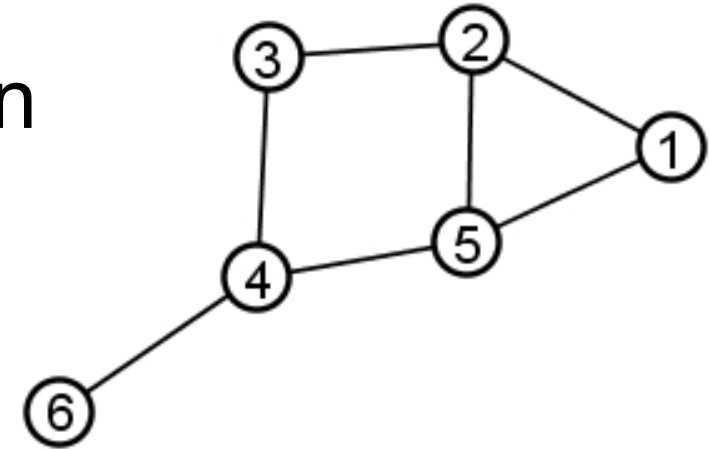
3. Impact on programming of on-line games

4. Other stories

A computer network is a system that connects two or more computing devices for transmitting and sharing information. Computing devices include everything from a mobile phone to a server. These devices are connected using physical wires such as fiber optics, but they can also be wireless.

Concepts

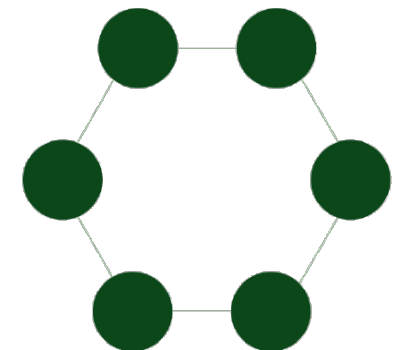
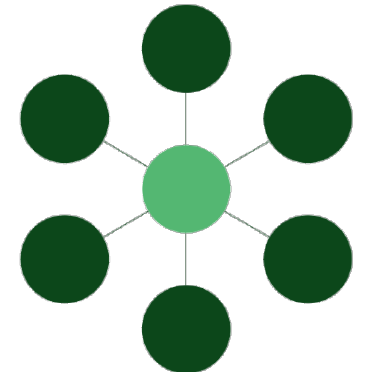
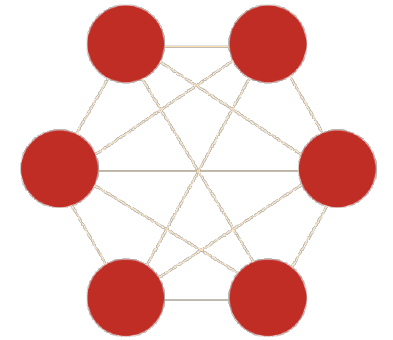
- **Node:** redistribution point or communication endpoint of a network
- **Link:** connection between two nodes
 - Unidirectional or Bidirectional
- **Topology:** arrangement of the nodes and links of a network
 - Physical or Logical
- **Route:** path from source to destination through a topology
- **Protocol:** set of rules that govern the communication between nodes in a network
- **Address:** identifier for a node in a network



“Protocols are to communication what algorithms are to computation”

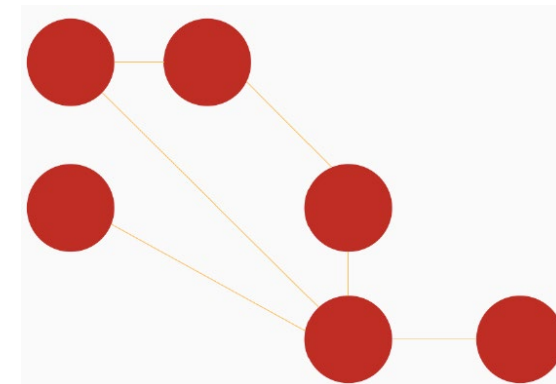
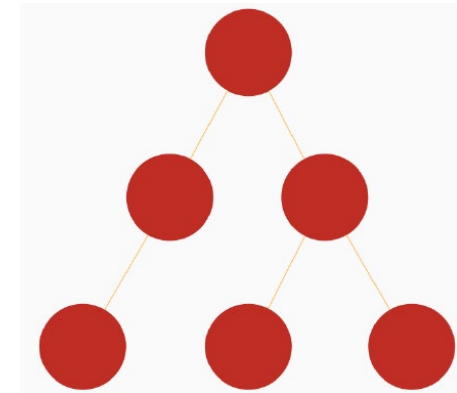
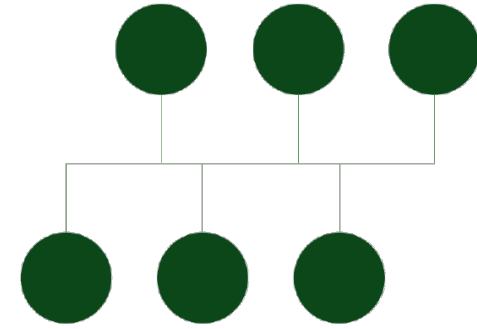
Topologies (I)

- **Fully connected:** each pair of nodes has a link that connects them. Adding a new node implies $N-1$ new links. *How many links for N nodes?*
- **Star:** each node is connected to a central switch or hub. The central node is in charge of all communications. *Resilience?*
- **Ring:** data travels along one direction until reaches the receiver. Each node receives, checks if it's for him or not, and passes. *Delay?*



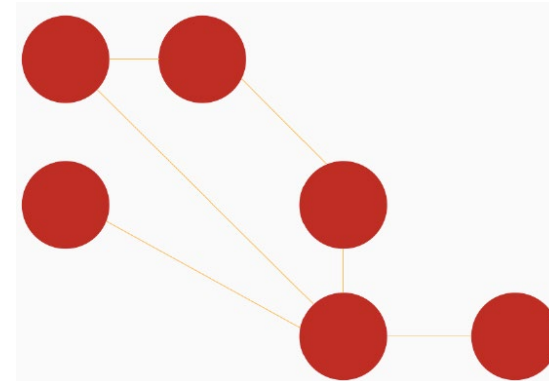
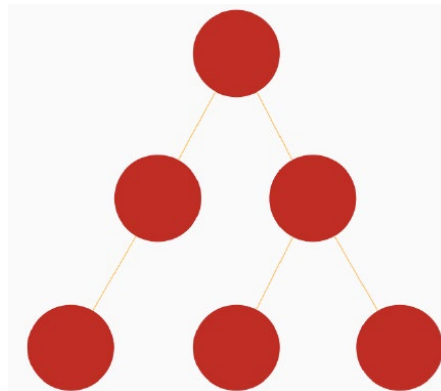
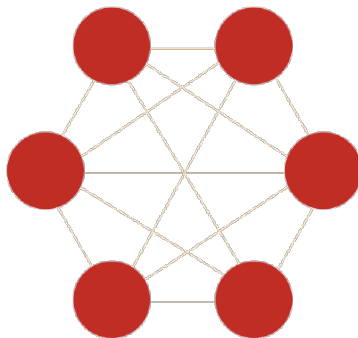
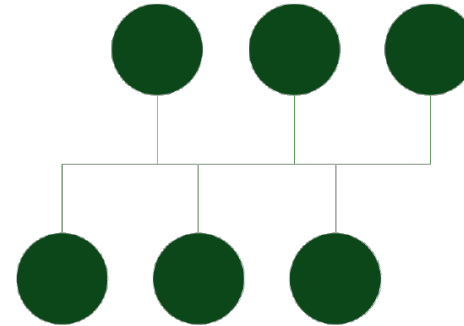
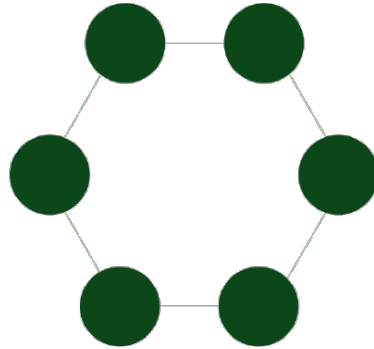
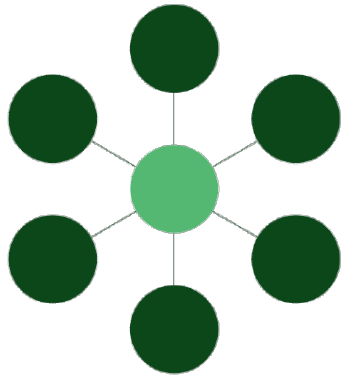
Topologies (II)

- **Bus:** each node is connected to a shared medium of transmission. Messages arrive to all the nodes connected to the bus. *Interference?*
- **Tree:** there is a hierarchy and the root is the origin. Nodes are connected to parents (above) or children (below), but never siblings (lateral)
- **Mesh:** nodes are partially connected to others with no predefined shape. All nodes cooperate in the dissemination of data.



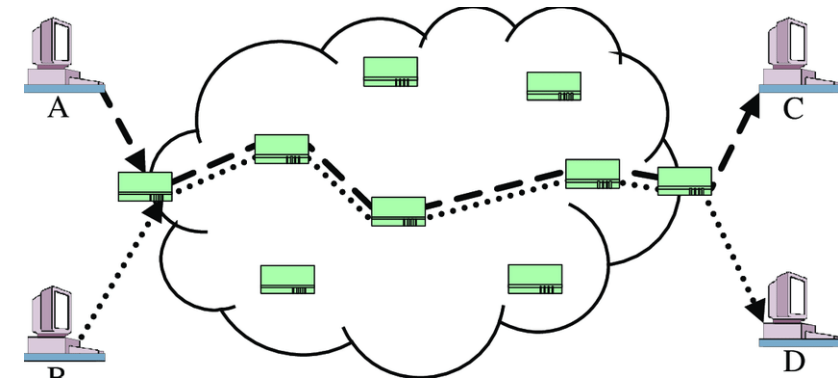
Exercise – Internet Topology

What topology you think best represents a network like the Internet?



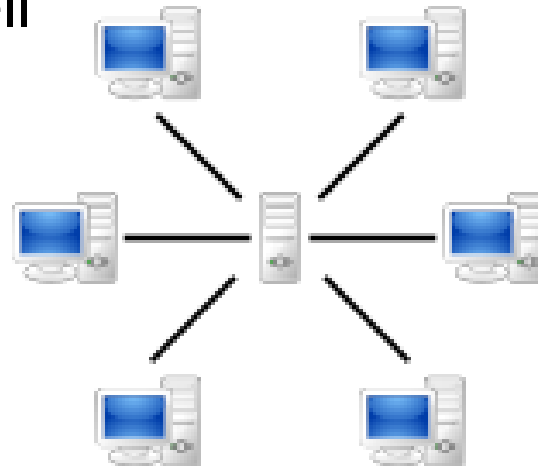
Mechanisms

- **Circuit switching:** the path between source and destination is reserved. When available, the transmission starts. When finished, the path is tore down. Problem: the lack of connections.
- **Packet-switching:** Each node transmits the messages in the form of small *packets* and only to the nodes it is connected. To reach the destination, routing is needed. Problem: congestion in certain links when there is a lot of traffic.



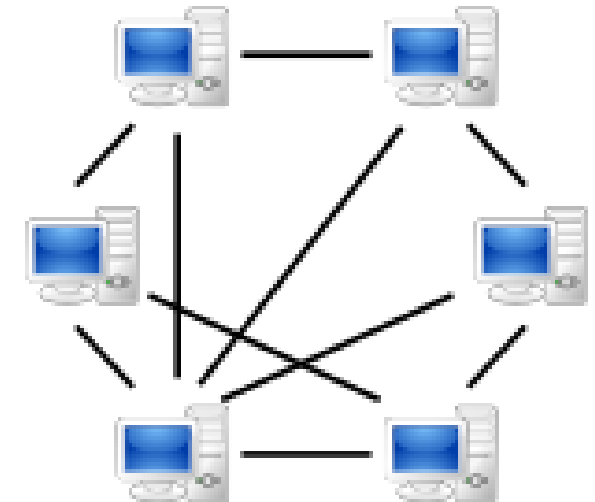
Client-Server Architecture

- Centralized
- The server is in charge of fulfilling requests from clients
- Logical topology: star
- Needs a high-performance and resilient (set of) servers to work
- Does not scale well



Peer-to-Peer (P2P) Architecture

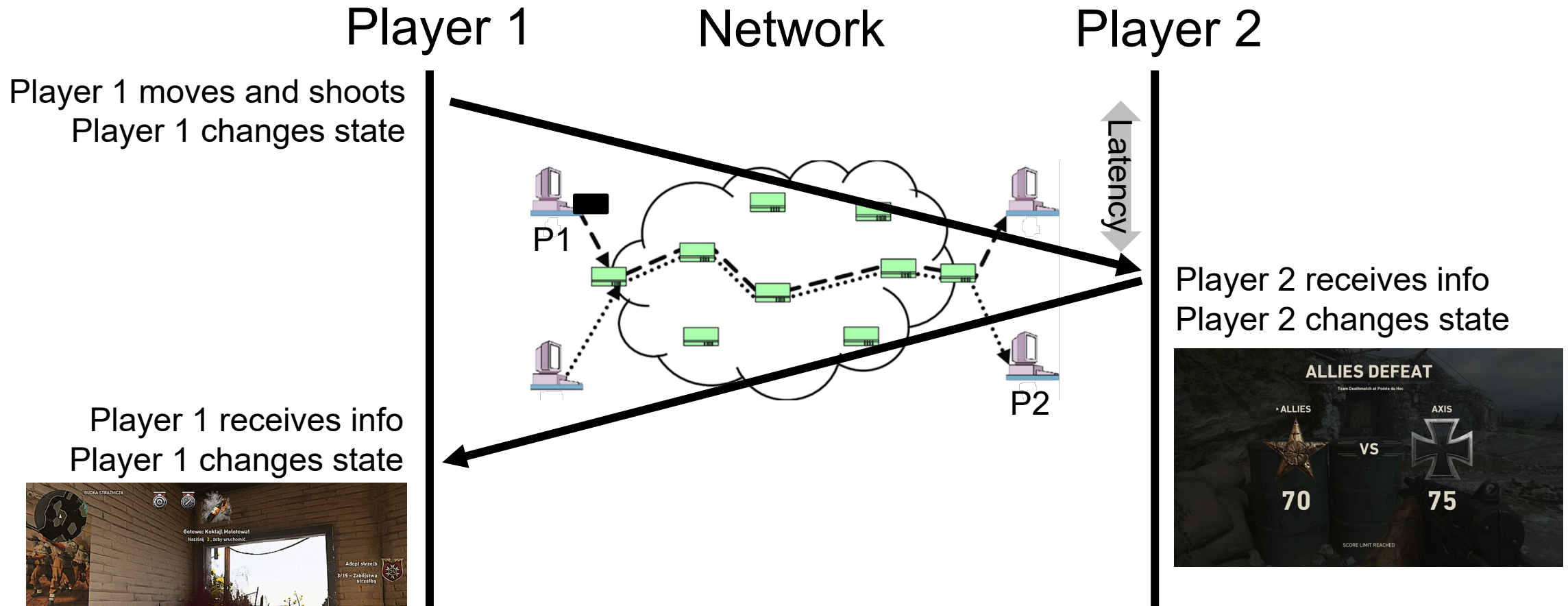
- Distributed
- All nodes are clients and servers, sharing responsibilities
- Logical topology: mesh
- Needs node discovery mechanisms and enhanced security to work
- Scalable



- **Latency:** (delay) time required to send a message from source to destination [sec]. Causes lag.
- **Jitter:** temporal variability or fluctuation of latency. How smooth is the latency statistic [sec]. Causes glitches.
- **Packet loss:** amount of messages lost in the network due to congestion, disconnected nodes, etc... [%]
- **Bandwidth:** maximum speed of transmission [bit/s]
- **Throughput:** rate of successful transmissions in a link or network [bit/s]

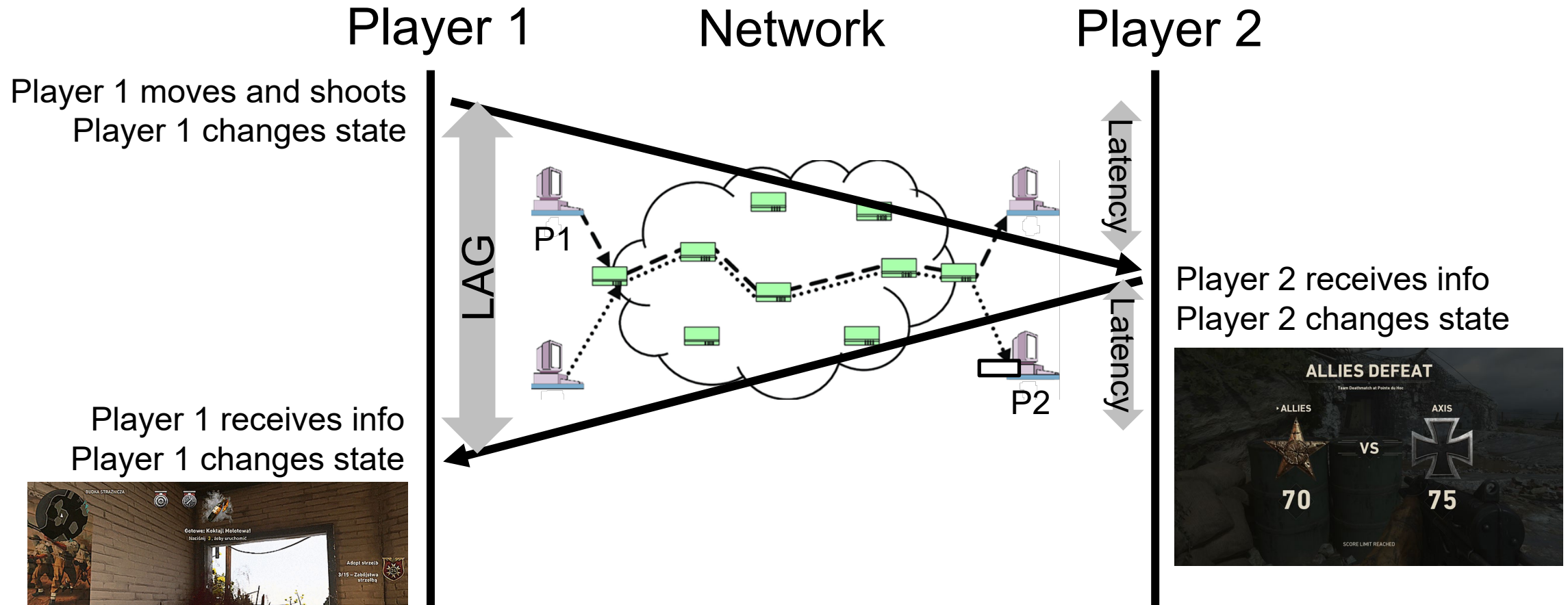
Metrics

- **Latency**: (delay) time required to send a message from source to destination [sec]. Causes lag.



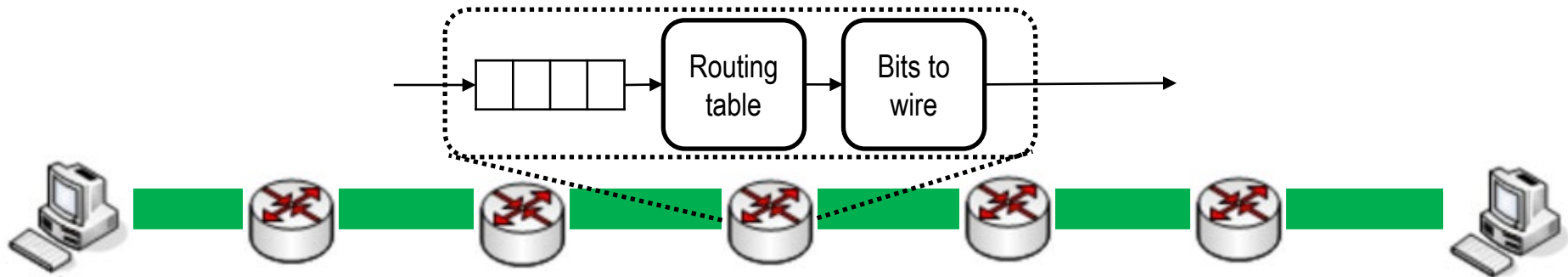
Metrics

- **Latency**: (delay) time required to send a message from source to destination [sec]. Causes lag.



Network Latency: Origins

- Network latency generally dominates over others.
- The main causes are:
 - **Multiple hops:** each router adds its own latency
 - **Queuing:** the router may not be ready and the packet is queued
 - **Processing:** check IP, calculate route, prepare to send
 - **Transmission:** time to put the bits into the physical medium
 - **Propagation:** the bits travel at the speed of light



Network Latency: Equations

- Network latency generally dominates over others.
- The main causes are:
 - **Multiple hops:** N_H
 - **Queuing:** T_Q independent on L
 - **Processing:** T_P independent on L
 - **Transmission:** $T_{TX} = \frac{L}{B}$
 - **Propagation:** $T_{PR} = \frac{D}{V}$

L is packet length (bits)
 B is link bandwidth (bps)
 D is physical distance (m)
 V is propagation speed (m/s)

Default for propagation speed
is the speed of light: $3 \cdot 10^8$ m/s

$$Latency = \sum_{N_H} (T_Q + T_P + T_{TX} + T_{PR})$$

Network Latency: Estimates



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- Network latency generally dominates over others.
- The main causes are:
 - **Multiple hops:** 1-100, you can check with a ping (TTL)
 - **Queuing:** unpredictable, depends on busy routers
 - **Processing:** generally small, a few ms.
 - **Transmission:** packets are maximum 1500 Bytes in length, 10-Mbps ethernet cable would take $1500 \cdot 8 / 10^7 = 1.2$ ms
 - **Propagation:** intercontinental cable of 6000 Km, the bits reach the other end in around $6 \cdot 10^6 / 3 \cdot 10^8 = 20$ ms. Try to have servers geographically close...

Exercise – Packet Size Estimation



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Take the example of a FPS game. What kind of information you think we would need to send to the network? Estimate the amount in a single frame.



Exercise – Packet Size Estimation



CITM

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```
struct playerState {  
    vec3f position;  
    vec3f linear_velocity;  
    quat4f orientation;  
};
```



Exercise – Packet Size Estimation



CITM

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```
struct playerState {  
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```

$$L = 3 * 32 + 3 * 32 + 4 * 32 = 320 \text{ bits}$$



Exercise – Packet Size Estimation



CITM

Take the example of a FPS game. What kind of information you think we would need to send to the network? Estimate the amount in a single frame.

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struct playerState {  
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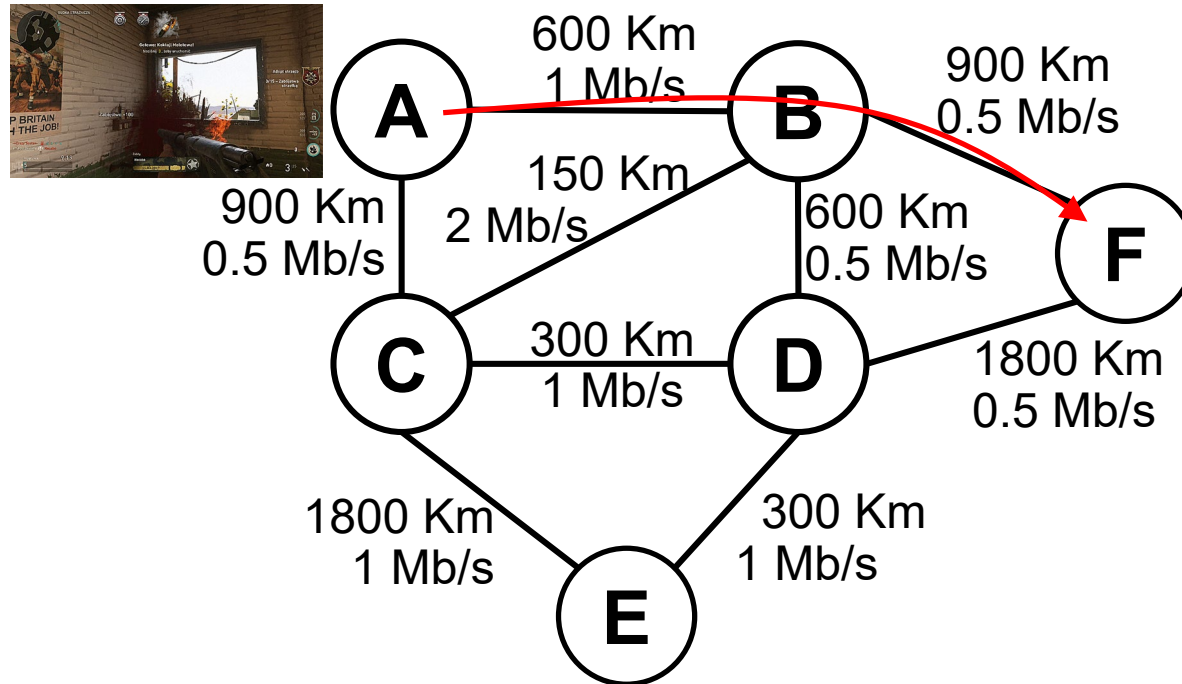
**This is one of many
possible implementations
depending on the game**

$$L = 3 * 32 + 3 * 32 + 4 * 32 = 320 \text{ bits}$$



Exercise – Latency Calculation

Consider that your 320-bit packet goes from you (node A) to the other player (node F) through the path ABF. How many hops does it take? Which is the latency? Assume a queuing latency of 1ms and a processing latency of 0.1 ms.



Exercise – Latency Calculation

Consider that your 320-bit packet goes from you (node A) to the other player (node F) through the path ABF. How many hops does it take? Which is the latency? Assume a queuing latency of 1ms and a processing latency of 0.1 ms.

- How many hops? **2**
- Which is the latency?

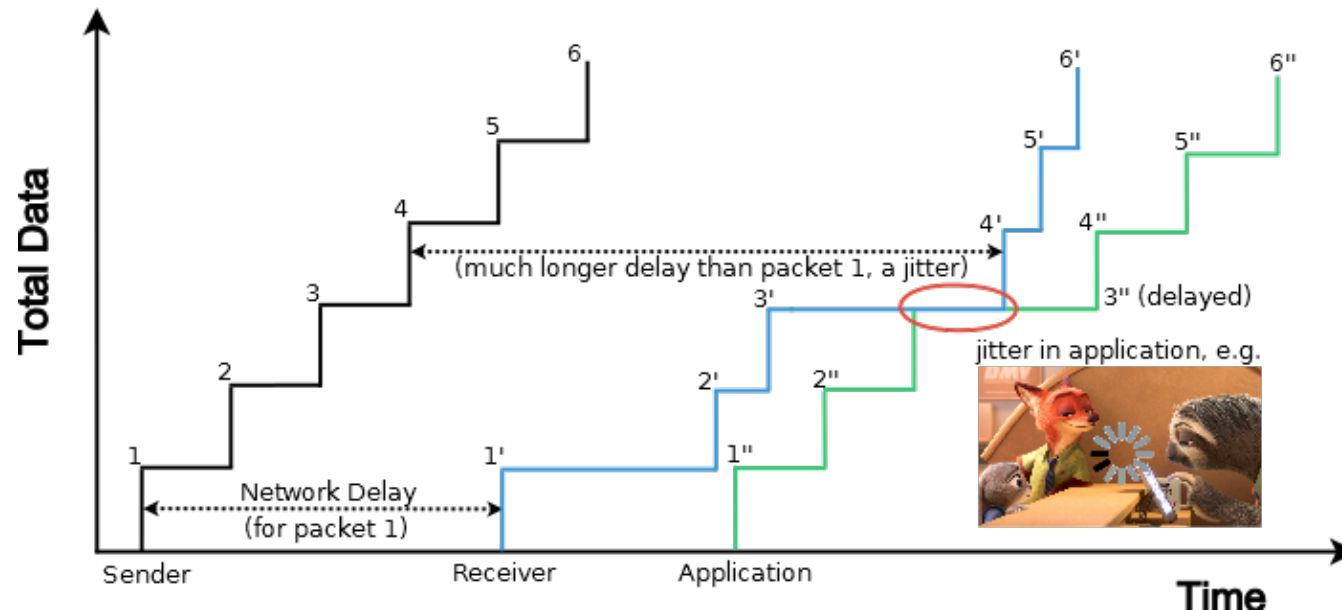
$$T_{TX_{AB}} = \frac{320 \text{ b}}{10^6 \text{ b/s}} = 0.32 \text{ ms}$$

$$T_{PR_{AB}} = \frac{6 \cdot 10^2 \text{ Km}}{3 \cdot 10^5 \text{ Km/s}} = 2 \cdot 10^{-3} \text{ s} = 2 \text{ ms}$$

(calculation omitted for BF)

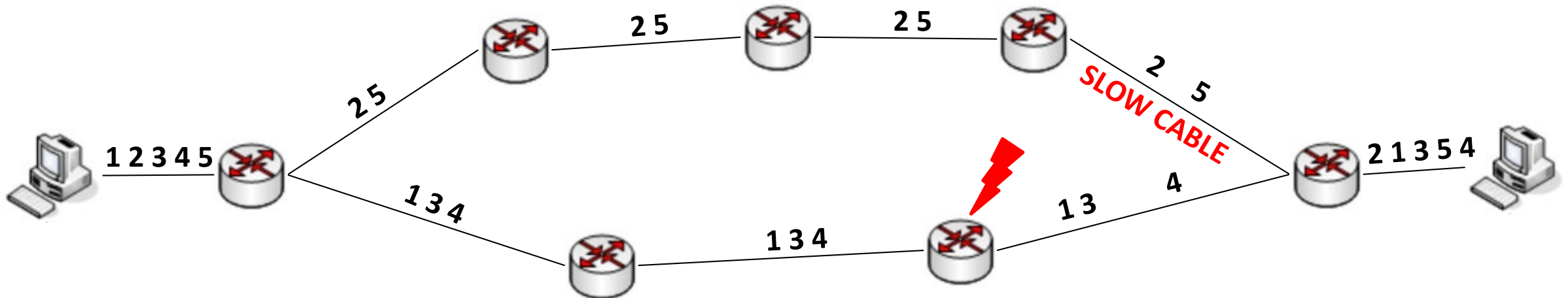
$$Lat_{AF} = Lat_{AB} + Lat_{BF} = (0.1 + 1 + 0.32 + 2) + (0.1 + 1 + 0.64 + 3) = \mathbf{8.16 \text{ ms}}$$

- **Latency:** (delay) time required to send a message from source to destination [sec]. Causes lag.
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Network Jitter: Origin

- Jitter is basically caused by the variable network delays experienced in different routers.
 - Distances and speeds may be different on different paths
 - Queuing delays cannot be predicted
- It can lead to disordered packets!



- Jitter J is calculated as the average deviation of the mean. One can find several formulas, but we will use:

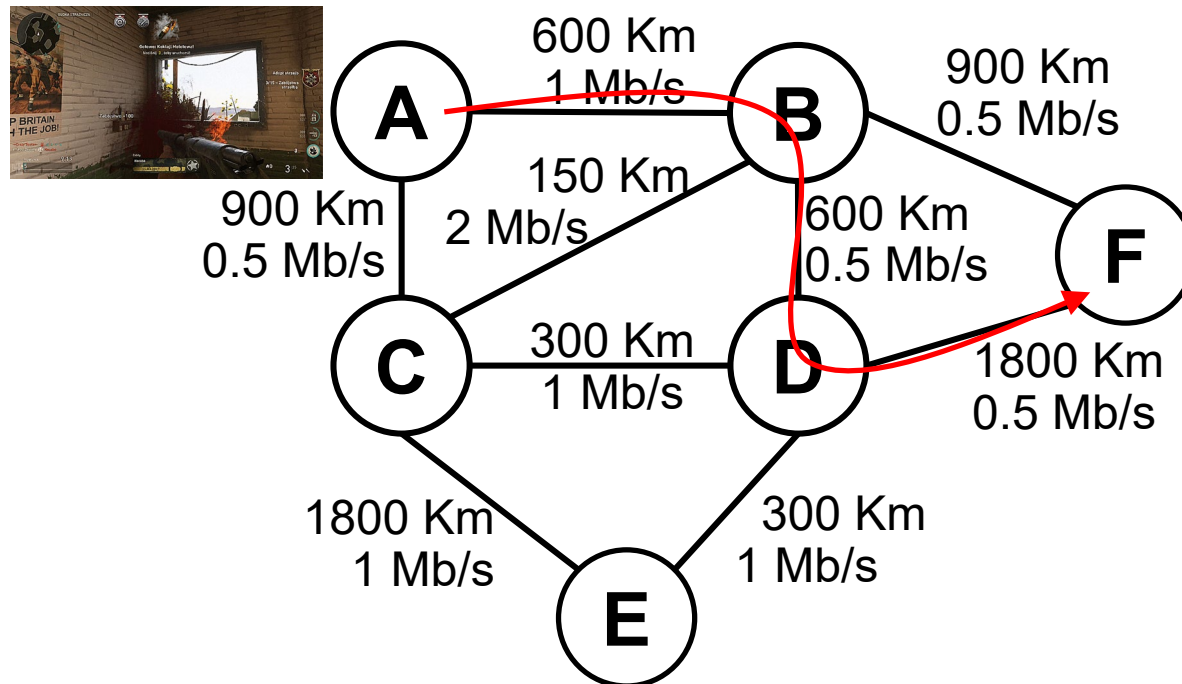
$$J = \sqrt{\frac{1}{N} \sum_{i=1}^N (Lat_i - \overline{Lat})^2}$$

Where:

- N is the number of measured packets
- Lat_i is the latency of packet i .
- $\overline{Lat}$ is the average latency across all measured packets

Exercise – Jitter Calculation

Consider a second packet that now needs to take a detour and goes through the path ABDF. How many hops does it take now? Which is the latency of this second packet? What is the jitter after these two packets?



Exercise – Jitter Calculation

Consider a second packet that now needs to take a detour and goes through the path ABDF.

- How many hops? **3**
- Latency? $Lat_{AF} = Lat_{AB} + Lat_{BD} + Lat_{DF} = (0.1 + 1 + 0.32 + 2) + (0.1 + 1 + 0.64 + 2) + (0.1 + 1 + 0.64 + 6) = 3.42 + 3.74 + 7.74 = \mathbf{14.9\ ms\ (vs\ 8.16\ ms)}$
- What is the jitter? $\overline{Lat} = \frac{(14.9+8.16)}{2} = 11.53\ ms$

$$J = \sqrt{1/2 [(14.9 - 11.53)^2 + (8.16 - 11.53)^2]} = 3.37\ ms$$



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- **Packet loss:** amount of messages lost in the network due to congestion, disconnected nodes, etc... [%]

- Packet loss is basically a side effect of using an unreliable communications medium.
- Causes for dropped packets:
 - **Unreliable physical medium:** badly shielded cables, interference in the WiFi. That makes bits to “flip” (0 becomes 1, or vice versa) and be seen as errors.
 - **Unreliable link:** when the WiFi is too congested and you cannot access your router, you will end up dropping the packet
 - **Unreliable network:** routers in the network have a finite queue. Buffer overruns are typical causes of packet drops.

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Exercise – Throughput Estimation



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Take the example of our FPS game before. Assume we need to send our status every single frame and we target 60 frames per second. Which is the throughput required to move my avatar?



Exercise – Throughput Estimation



CITM

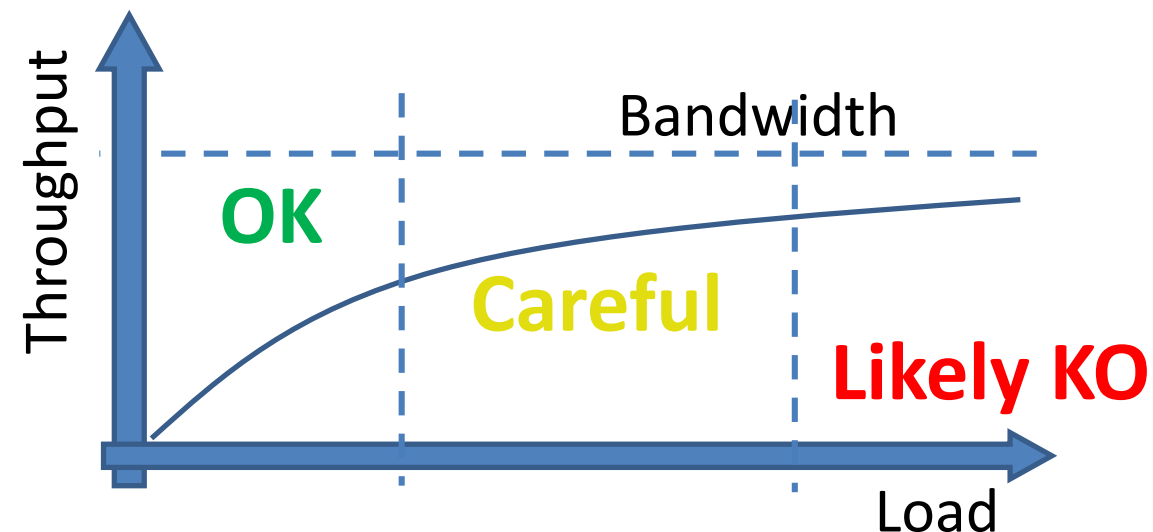
Take the example of our FPS game before. Assume we need to send our status every single frame and we target 60 frames per second. Which is the throughput required to move my avatar?

$$Thr = 320 \frac{bits}{frame} * 60 \frac{frames}{second} = 19.2 Kbit/s$$



Bandwidth and Throughput

- Throughput is by definition lower than bandwidth, that is the maximum speed possible
- It is better that your game requires a throughput (load) much lower than the user's bandwidth to avoid problems
 - Our example is like that: 19.2 Kb/s against 100 Mb/s of a regular Internet connection. But...
 - We might want to add more players, objects, etc.
 - We will see that the netcode and the network imply overheads
 - The network loses packets so we will need retries



Conclusions



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- The Internet is a large network with many nodes, links, topologies and architectures.
- It is unreliable and adds latency, jitter, and packet losses. Packets can even arrive out of order!
- These will cause lag and glitches in your game unless we remedy it.



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Others

- Local Area Network (LAN)
vs
Wide Area Network (WAN)

